waters, mangrove swamps and coastal lagoons. Estuarine fish are more common in Bengal and Kerala. Culturing of animals in the water having salinity range 0.5 – 30 ppt are called as brackish water culture. Fishes cultured in brackish water are Milk fish (*Chanos Chanos)*, Sea bass ('Koduva'), Grey mullet ('Madavai'), *Pearl spots ('Kari'meen)* etc.,

Marine Fisheries deal with fishing operations along seacoasts. The Indian subcontinent approximately has a 5600 kms long coastline. About 80% of India's marine fish are supplied by the west coast and the remaining 20% by the east coast. The premier varieties are mackerels, sardines, sharks, and catfish. Marine edible fishes of Tamilnadu coast include both cartilaginous and bony fishes. Culturing of animals in the water salinity ranges from 30 - 35 ppt is called Mariculture. Some fishes like *Chanos* sp, *Mugil cephalus* are cultured here. Culturing of animals in the salinity ranges from 36 - 40 ppt is called Metahaline culture. Eg, Brine shrimp (*Artemia salina)*. Artemia is commonly known as the brine shrimp. It is a crustacean and lives in high saline waters because of its high osmoregulatory capacity.

13.7.1 Fish culture

Characteristics of cultivable fishes

The special characteristic features of cultivable fishes are:

i. Fishes should have high growth rate in short period for culture.
ii. They should accept supplementary diet.
iii. They should be hardy enough to resist some common diseases and infection of parasites.
iv. Fishes proposed for polyculture should be able to live together without interfering or attacking other fishes.
v. They should have high conversion efficiency so that they can effectively utilize the food.

Types of cultivable fish

Cultivable fish are of 3 types(Figure 13. 8).

a. Indigenous or native fresh water fishes (Major carps, *Catla*, *Labeo*, *Clarias*)
b. Salt water fishes acclimatized for fresh water (*Chanos*, Mullet).
c. Exotic fishes or imported from other counties (Common carps)

Among these, major carps have proved to be best suited for culture in India, because the carps

1. Feed on zooplanktons and phytoplanktons, decaying weeds, debris and other aquatic plants.
2. They can survive in turbid water with slightly higher temperature
3. Can tolerate O_2 variations in water.
4. Can be transported from one place to other easily.
5. They are highly nutritive and palatable.

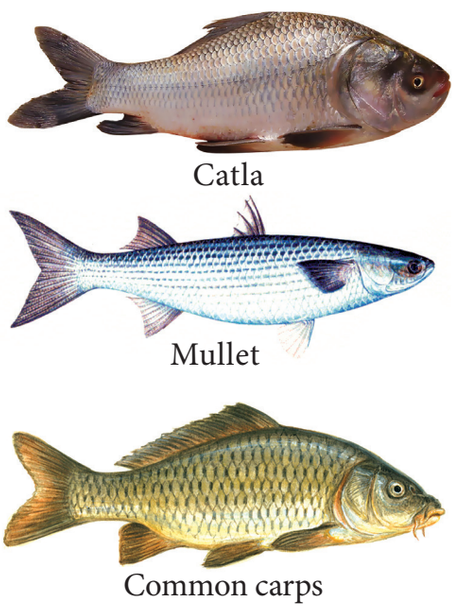


Figure 13.8 Different types of freshwater cultivable fishes

XI Std Zoology Chapter-13 EM.indd 250 30/12/2022 14:08:33